

HOMEPLOTSAPPSLIVE EDITORINSERTVIEW

File Edit View Tools

MATLAB Drive

Workspace

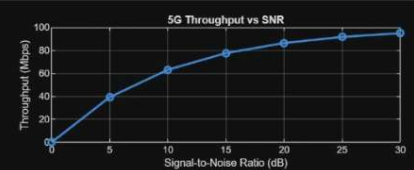
Name	Value
Max_TP	100
SNR	[0,5,10,15,20]
Through...	[0,39.3469,6...

5g and 6g exp 7.1.mlx

untitled * x

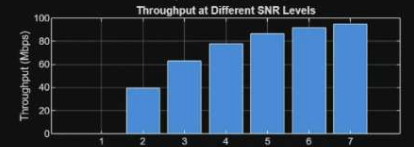
```
5 SNR = 0:5:30;
6 % Maximum Throughput (Mbps)
7 Max_TP = 100;
8 % Throughput calculation
9 Throughput = Max_TP*(1-exp(-SNR/10));
10 disp('SNR(dB) Throughput(Mbps)')
11 disp([SNR' Throughput'])
12 figure
13 % ----- Graph 1 -----
14 subplot(2,1,1)
15 plot(SNR,Throughput,'-o','LineWidth',2)
16 title('5G Throughput vs SNR')
17 xlabel('Signal-to-Noise Ratio (dB)')
18 ylabel('Throughput (Mbps)')
19 grid on
20 % ----- Graph 2 -----
21 subplot(2,1,2)
22 bar(Throughput)
23 title('Throughput at Different SNR Levels')
24 xlabel('SNR Index')
25 ylabel('Throughput (Mbps)')
26 grid on
```

5G Throughput vs SNR



SNR (dB)	Throughput (Mbps)
0	0
5	39.35
10	63.21
15	77.88
20	86.47
25	90.48
30	93.73

Throughput at Different SNR Levels



SNR Index	Throughput (Mbps)
1	39.35
2	63.21
3	77.88
4	86.47
5	90.48
6	93.73
7	95.00

Command Window

New to MATLAB? See resources for [Getting Started.](#)

HOMEPLOTSAPPSLIVE EDITORINSERTVIEW

NewOpenSaveExportCompareGo ToFindBookmark

FILE

NAVIGATE

TEXT

CODE

ANALYZE

TEST

SECTION

RUN

pranay

Workspace

Name	Value
BER	[0.3000,0.1800,0.1000,0.0500,0.0200,0.0080,0.0010]
Reliability	[70,82,90,95,98,99,100]
SNR	[0,5,10,15,20,25,30]

5g and 6g exp 7.1.mlxuntitled * x

```
5 SNR = 0:5:30;
6 % Simulated Bit Error Rate (BER)
7 BER = [0.30 0.18 0.10 0.05 0.02 0.008 0.001];
8 % Reliability (%)
9 Reliability = (1-BER)*100;
10 disp('SNR(dB) BER Reliability(%)')
11 disp([SNR' BER' Reliability'])
12 figure
13 % ----- Graph 1 -----
14 subplot(2,1,1)
15 semilogy(SNR,BER,'-o','linewidth',2)
16 title('SNR vs Bit Error Rate in 5G')
17 xlabel('Signal-to-Noise Ratio (dB)')
18 ylabel('Bit Error Rate (BER)')
19 grid on
20 % ----- Graph 2 -----
21 subplot(2,1,2)
22 plot(SNR,Reliability,'-s','linewidth',2)
23 title('SNR vs Communication Reliability')
24 xlabel('Signal-to-Noise Ratio (dB)')
25 ylabel('Reliability (%)')
26 grid on
```

Command Window

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>>

Files

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